Chapter 5: Food, Glorious Food

Food is the source of all the nutrients our bodies need to stay forever young and fit. In a world where food is often demonized as the cause of obesity and illness, it is easy to forget that we actually need to eat well to stay alive. This chapter is extensive because lots of facts about nutrition need clarification. To be able to eat right, we must know nutritional science.

We shall focus on the importance of getting the right nutrients from the food we eat by

- discussing the macro and micronutrients and their role in building, maintaining, and replenishing physiological structure and function;
- laying to rest common diet myths;
- doing away with fad diets;
- doing away with calorie counting and starvation; and
- learning how to make good nutrition a natural lifestyle choice.

What is food really? Everything we eat, be it the finest-quality filet steak or a hamburger at a fast-food joint, ultimately gets broken down into its basic building blocks in our digestive system. The nutrients derived from digestion can be very simply categorized by the relative quantities that exist in nature—the macronutrients and micronutrients.

Macronutrients, or "macros," make up the bulk of the dry weight in foods. These are all organic in nature and include carbohydrates, proteins, and fats. Their molecules are large, complex building blocks, which are materially essential for the structure and form of all living organisms. Carbohydrates and fats are also the main energy source for living cells.

Our bodies need micronutrients, or "micros," in small quantities. These are the important substances that are necessary for biochemical reactions to take place effectively within living cells. Micronutrients include organic compounds such as vitamins, phytochemicals, and inorganic compounds like mineral salts.

Organs, Tissues and Cells

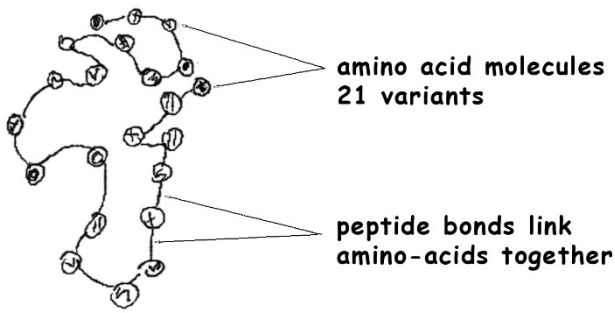
The body is made up of organs like the brain, heart, lungs, skin and so forth. These organs are made up of tissues like muscle tissues and liver tissues, or the connective tissues that encase the organs keeping them in shape. These tissues are made up of specialized cells. For example liver cells are found in liver tissue, and muscle fibers are specialized cells only found in the muscles. The cells are units of life. These cells are made of proteins, fat and some components of carbohydrates. The cell are also connected together by intercellular material, which is made up of mostly protein.
Cells, being living units, need constant supply of nutrients for function and energy. Cells also die and get replaced. These processes require nutrients that have to come from our diet.

Protein Builds Life

Our lean body mass (mass of the body excluding the excess fat mass) contains about 21 percent protein. Protein can be considered the building block of life. The function of protein in every living organism is the following:

- Structure. We all know protein is an important component of muscle cells and hence movement. Protein is also the structural component of tissues in the body—from the tendons that link the muscles to bones; to connective tissues that hold the organs in shape; to collagen that make up skin; to hair, nails, and even parts of bone.
- Cell (biochemical) function. Smaller protein molecules such as enzymes, hormones, organic cofactors, cell receptors, neurotransmitters, carriers, and biochemically active compounds are important molecules that help along all the biochemical processes in the cells of the body.
- Viscosity of blood and fluid in cells.

Protein molecules are chains of amino-acids linked together by peptide bonds. The unique sequence of amino-acids create the specific protein structure.

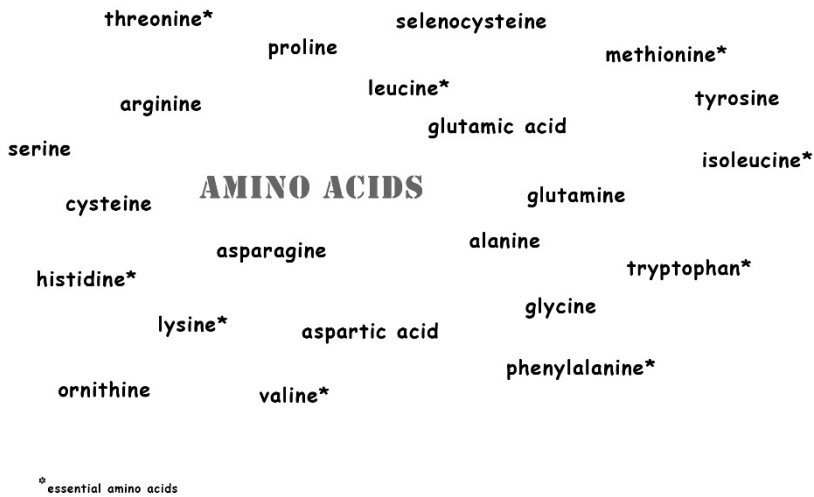


Proteins are large molecules made up of building blocks called amino acids. Combinations of twenty-one amino acids make up all the proteins in the human body. Of these twenty-one amino acids, twelve can be produced by the body itself, and nine are not produced by the body and must come from our food source.

These nine amino acids are called "essential amino acids." Amino acids, especially the nine essential amino acids, are important components for a healthy, complete diet. Amino acids are building units of protein for the structure and function of cells in tissues and organs. Cells in our bodies make the required protein according to the information and signals given by genetic material (DNA). Each protein molecule must have the amino acids strung up in an exact combination to be functional.

Amino acids also have complex roles as neurotransmitters (the chemicals that send messages between nerve cells) and help in building up other important biological compounds. Long-term deficiency in amino acids will result in the degeneration of health and accelerated aging.

Plants and animal sources of food contain protein. The most complete sources of amino acids, however, come from animal protein: meat, poultry, fish, and eggs. Animal products are easily digestible, and animal protein contains all the amino acids necessary for healthy body function.



Beyond the nutritional benefits of consuming adequate amounts of protein in our diets, there is an added plus to eating protein at mealtime. That is obesity control. Proteins take more energy to digest than refined carbohydrates, sugar, and alcohol do. Proteins in the diet contribute less to net calorie input than carbohydrates do.

Proteins also tend to stay in the stomach longer than refined carbohydrates. Research has shown that proteins leave us feeling satisfied faster and longer, so we're less likely to snack or even overeat.

Proteins in the diet don't induce insulin surges we get after a carbohydrate-rich meal, which means that there is no likelihood of excessive fat accumulation and no hindrance to fat burning. There is more discussion about insulin effects in the section on carbohydrate metabolism in this chapter.

Plan Your Meals Around a Protein Source

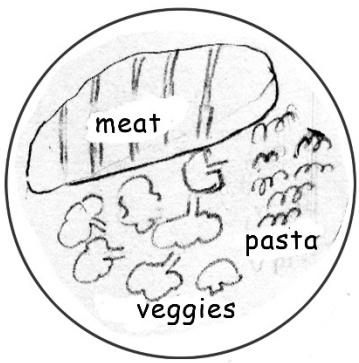
There is no hard-and-fast rule as to how much protein we should consume at each meal. We should not get number bound and start counting and weighing. The easiest rule to follow is that the protein component of food should take priority. Make sure there is enough protein on your plate. Even when you snack, make sure there is protein in the snack.

Another strategy to get more protein into your diet is to eat the protein component first. If you eat all the protein on your plate first,

there will be less space in the stomach for nutrient-poor carbohydrates that might be there.

Plan your meals around protein. Good protein sources in common foods include the following:

- Eggs
- Poultry
- Lean cuts of beef
- Lean cuts of pork
- Fish (oily fish is actually good for you)
- Shellfish
- Protein-rich vegetables like legumes and nuts



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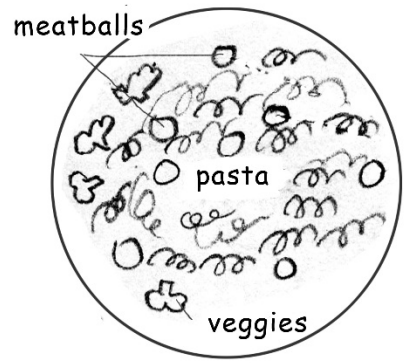

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Protein, not carbs, should be main part of the dish.

The above protein-rich whole foods should make up the bulk of your meal. This means you have enough of these protein sources to satisfy hunger. Have the vegetables, fats, and carbohydrates on the side. Planning meals this way is the secret to getting enough proteins for prolonged health and fitness.

Fat is Not the Enemy

The human body has two categories of fat:

- Storage fat. Storage fat is the excess fat stored under the skin in tissues called adipose tissues. This excess fat is harmless and

needed during Paleolithic (stone-age) times to survive famine. Nowadays, of course, fat in the adipose tissues is regarded as a nuisance because it makes us look flabby, something to be "burned off." This is the fat we understand as the indicator of obesity. Depending on individuals, the percentage of storage fat varies significantly.

- Essential fat. Essential fat is necessary fat for human survival.

Essential fat percentage in men is 7 percent and in women is 12 percent.

What is the function of fat in the body? Every cell in the body is made up of at least 2 percent fat. Nerve and brain cells contain more fat. Fats in the cells and body fluids are called "lipids." Lipids make up the cell structure. Without lipids, the cell would not exist.

- Fat is an essential stored-energy source. If the diet is right, fat is the body's preferred source of energy.
- Some internal organs are protected by fatty tissue, which acts as a cushion and support.
- Fat is a carrier to transport essential chemicals and hormones for biological function.
- Fat is a lubricant in organs.
- Free fatty acids in cells and the bloodstream are used as fuel to build and modify protein and for producing other compounds like neurotransmitters (molecules responsible for nerve function).

When we take fat out of our food, we become deficient in the following nutrients:

- Essential fatty acids like omega-3 fatty acids and omega-6 fatty acids. These are building blocks of cells and factors needed for cell function.
- Vitamins like vitamins A, D, E, and K which can only dissolve in fat (fat-soluble vitamins) and other lipid-soluble compounds. These vitamins need fat, not water, as a solvent so they can be transported into the blood. Low-fat or non-fat food lack these vitamins.

A diet deficient in fat means a deficiency in essential nutrients. This leads to degeneration and deterioration of the body, which is a primer for accelerated aging and chronic illnesses.

We need fat in our diets. Good fat sources include olive oil, fish, nuts and seeds, avocado, coconut, and eggs.

Eating Fat Alone Will Not Make You Fat

Refined carbohydrates, including sugar, in our diets cause obesity, not dietary fat. Processed, low-fat foods are no healthy option!

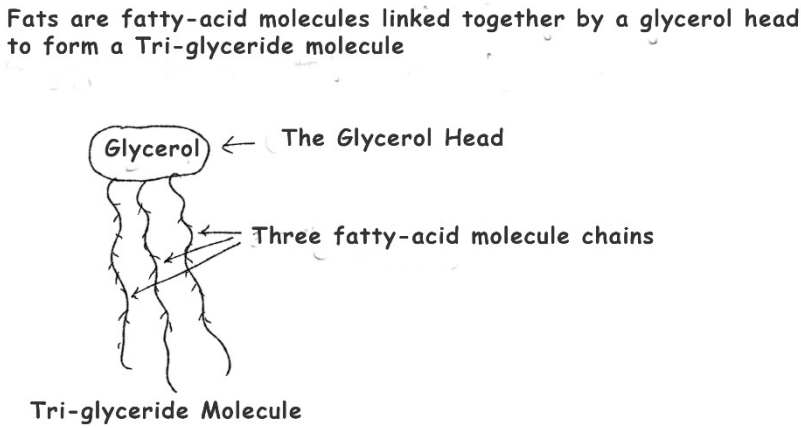
Those who cook often understand the importance of fat to the taste and look of cooked food.

A home-baked pound cake, for example, made without butter will turn out crumbly, dry, and rock hard. If you try to make nonfat yogurt with skimmed milk, you'll get an unappetizing, lumpy product. Fat makes food delicious because it is a carrier for aromatic molecules that make food smell and taste great. Fat also gives food its creamy sensation in the mouth by lingering on the tongue.

To create their factory-made "fat-free" versions of food, the food industry replaces fat with refined starches and gels. This step compensates for the loss in the appeal of the food when fat is absent. When you substitute your full-fat foods for the low-fat variety, you actually increase your refined carbohydrate intake.

Diets dominated by refined carbohydrates, as we will discuss later, are a cause of obesity, aging, and age-related illnesses. Fat, on the other hand, is an important nutrient.

There is really no good reason to choose low-fat versions of packaged foods over regular, full-fat foods.



Good Fat vs. Bad Fat

Foods that contain good fats are rich in essential fatty acids like omega-3 fatty acids. They also contain fat-soluble vitamins. Essential fatty acids are, as the term implies, important for a wide range of biological functions.

There is also such a thing as "bad fat" or unhealthy fat. These are the fats to avoid. These are trans fats.

Trans fats are human-engineered fats that can be found in highly processed baked, roasted foods; fried foods; and hydrogenated cooking oil. These fats are engineered to be more heat stable, enabling the food to be cooked more quickly. And since these fats are almost indestructible, they give processed foods a longer shelf life.

Trans fats can be found in meats but in much smaller quantities than in processed foods.

Trans fats wreak havoc upon entering the bloodstream, causing inflammation, clogging arteries, and leading to common chronic ailments.

The good news is that the world is slowly coming to grips with the harmful effects of trans fats. More governments are getting tough on its use in food.

As far as fat in your diet is concerned, it is better to choose whole

unprocessed foods that are rich in natural, good fats. These foods contain essential nutrients that are present in the fat necessary for health. Here are some healthy tips:

- Always have olive oil, nuts, seeds, and avocado available in your kitchen.
- Eat oily fish, meats, and eggs.
- Choose whole-fat dairy rather than the low-fat option.
- Avoid processed, baked, fried, and roasted snacks as well as margarine and spreads that may contain trans fat.

If you have been avoiding all fat in your diet, it is time to change. Replace refined carbohydrates in your diet with food that is rich in protein and good fat. Eat the "good" fats without fear. You need essential fatty acids for cell function. Cells are the basic units of tissues and organs.

Having enough good fat sources in your diet will supply the body with all the essential fatty acids and vitamins you need. This is a positive way to help you stay young and healthy.

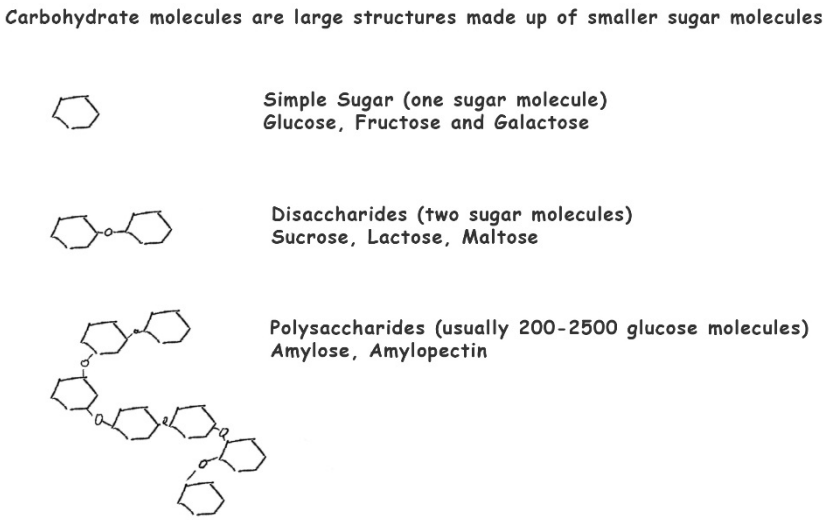
Carbohydrates: The Good and the Ugly

Carbohydrates are a source of energy for the human body. We consume them from plant sources and milk. Meats and eggs don't contain significant amounts of carbohydrate.

Carbohydrates are large molecules of reducing sugar units linked together. There are three types of reducing sugars naturally found in our diets:

- Glucose is the most common one and a component of all starches. Glucose is the universal fuel every cell in the human body uses. Because of its importance in metabolism, the glucose level in the blood is kept at a constant level by the hormone insulin. This is a very significant fact, which we shall discuss further in this chapter.
- Fructose is found naturally in plants. Due to its taste and sweetness, fructose is also used as an additive in our food and drinks. Table sugar, or sucrose, and high-fructose corn syrup (HFCS) are high sources of fructose in our diets. While fructose is a fuel source, overconsumption of this sugar has harmful effects on our health.

- Galactose is found naturally in milk and some plants. It is most commonly found in our diets as lactose.



Carbohydrates in the digested form enter the bloodstream as simple sugars. Disaccharides like sucrose break down into glucose and fructose, while polysaccharides break down into glucose molecules.

Simple sugars are single molecules of sugar. These molecules, which can move freely in the body via blood and other intracellular fluids, are also highly reactive. While being useful for cell metabolism in regulated quantities, they can cause damage to organs and cell structures.

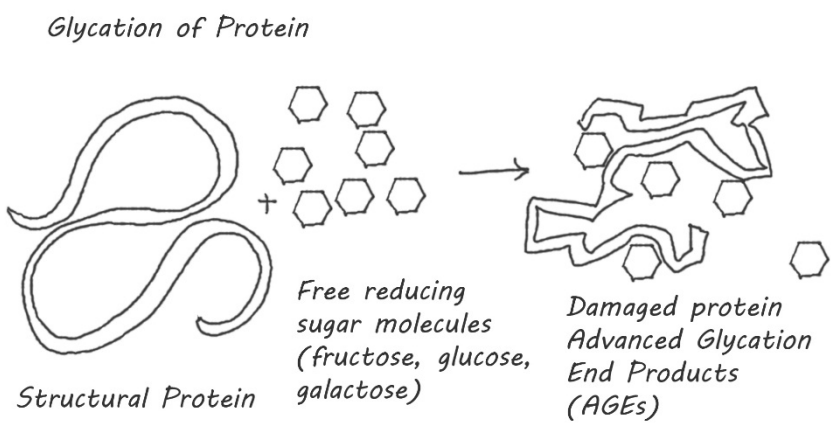
Excess glucose is removed through a regulatory mechanism that involves the hormone insulin. Where possible, glucose in the blood gets taken up by the cells in the body for metabolism, a process in which glucose molecules is broken down to carbon dioxide and water to produce energy. Excess glucose gets stored in the liver and muscles as larger carbohydrate molecules called glycogen. The glycogen storage in the human body is very limited. Further excess of glucose gets converted to fat by the liver. This fat is transported to the adipose tissues for storage.

There is no biological regulation in the body for fructose and galactose. These sugars are removed from the bloodstream only via the liver.

The Insulin Connection

Insulin is a hormone that functions to regulate the concentration of glucose in the blood and body fluids. While glucose is a fuel source for all cells in the body, it is also a harmful substance when consumed in excess. We must therefore keep the level of blood glucose at a constant level.

Glucose, fructose, and galactose are classified as reducing sugars. Reducing sugars are highly reactive molecules that bond spontaneously with protein and lipid molecules in a chemical reaction called glycation, which causes damage to the larger molecules. The resulting damaged protein or fat molecule is called advanced glycation end products (AGEs). The consequence of this damage is the breakdown in structure of cells and tissue.

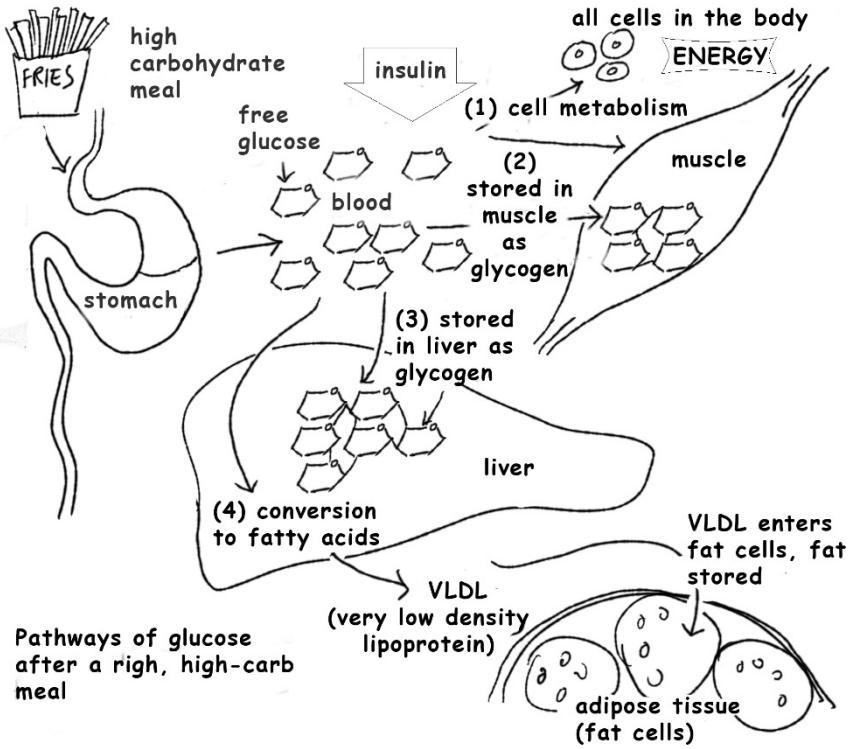


The toxic effects of glucose can commonly be seen in sufferers of untreated diabetes. The blood glucose levels in these individuals are constantly high. Free glucose molecules in high concentrations attack cellular structures, causing organ damage. Symptoms include blindness, neurological problems, stroke, and heart disease.

Glucose enters the bloodstream from the intestine as a result of digested carbohydrate and sugar. An increase in glucose concentration in the blood triggers insulin production. Insulin acts to mop up excess glucose by a number of processes:

- Storing glucose in the liver and muscle as glycogen. There is a limit as to how much glycogen can be stored. In general it is about five hundred grams in a standard human body.

- Converting excess glucose to fat
- Suppressing fat from being metabolized. In other words, fat burning is hindered since glucose in the blood must be used first.



Pathways of glucose after a righ, high-carb meal

There is a dangerous limit. The modern diet is high in refined carbohydrates. Every time we have a meal loaded with carbohydrates, blood glucose concentration goes up and insulin levels spike. Insulin is like a signaling device that activates different cells in the body to mop up excess glucose in the blood.

Constant insulin secretion has these consequences:

- Hunger swings. Hypoglycemia is a situation where blood-glucose levels dive to an uncomfortable low. When this

happens—usually less than two hours after a carbohydrate-loaded meal—we get tired and hungry. Hence the hunger swings, the carbohydrate cravings, and the vicious cycle of overeating.

There is a reason why carbohydrate-rich foods are highly addictive. Carbohydrate-laden foods affect pleasure sensors in the brain. Different carbohydrate sources affect the brain differently. That is why our "comfort food" lists commonly consist of these high-carbohydrate sources.

- Fat storing. Unlike plants, the human body has only limited capacity to store carbohydrates. Excess sugar in the blood has to be converted to fat for storage in the adipose tissues. The consequence is, of course, obesity.
- Fat sparing. Consistently high blood sugar and constant insulin spikes cause fat not only to build up but also to be locked in. Insulin signals all the cells in the body to prefer glucose as fuel since it has become readily available in the blood. As a result, the machinery for fat burning slows. This leads to fat build up and the difficulty to lose fat even with exercise.

When we're in the habit of consuming consistently high amounts of refined carbohydrates, our bodies learn to burn glucose as a main source of energy. We get hungry when the available glucose runs low. This happens very quickly since glucose doesn't stay in the blood of a non-diabetic person for very long.

A body that is running predominantly on glucose and is constantly secreting insulin will store up fat because fat is not its fuel of choice.

When the body is unable to use its fat stores, it runs out of energy very quickly, since insulin keeps glucose at a low level. This leaves the person feeling fatigued, lacking in energy, and in need of eating more carbohydrate-rich food.

It is a vicious cycle. The only way to break the cycle is to cut out refined carbohydrates as the main source of food—or cut it out completely, if possible.

The eventual system breakdown in the system caused by over consuming refined carbohydrates results in insulin resistance or type 2 diabetes. Cells in the body will cease to respond to insulin. This makes insulin function useless.

Those who don't suffer from diabetes might see the effects of overconsumption of carbohydrate-rich foods manifest as the following:

- Obesity

- Premature signs of aging
- Hypertension
- Cardiovascular disease
- Cancers
- Nerve- and muscle-coordination problems

We can reduce and eliminate these health risks by setting our minds on fully understanding carbohydrates in our diets.

Bad Carbohydrates, Good Carbohydrates and the Glycemic Index

Not all carbohydrate sources are alike. There are the bad and the good. Bad carbohydrate sources include foods that cause blood sugar levels in the blood to rise quickly on digestion.

The effect of bad sources of carbohydrates includes the following:

- They cause a sudden increase in blood sugar. This rise forces insulin to spike, contributing to a chain of biochemical reactions in the body that are responsible for obesity.
- They contain no or very little micronutrients. Consuming these sources of carbohydrates puts excess energy into our bodies without providing any other essential nutrients. It is what we call "empty calories." Excess energy converts to fat.
- They lead to the consumption of empty calories, resulting in a condition of malnourishment accompanied by obesity. Malnourishment leads to the degenerative process of aging. This is the reason why someone can get fat while the organs in the body starve until they become sick.
- They contain no or very little fiber. Fiber slows down the absorption of glucose into the blood, keeping its concentration at a safe level. The only way to get enough fiber is to consume whole vegetables and fruit.

We mostly find examples of bad carbohydrate sources in processed food, including the following:

- Refined flour-based foods like breads, cakes, pastries, and biscuits, including their whole-grain versions
- Sauces and dressings containing starch, flour, and sugar
- Jams and preserves
- Candy or confectionery

- Soft drinks, fruit juices, and sugared beverages
- Noodles and pasta
- Sweet desserts
- Starchy vegetables like potatoes
- Grains like rice, oats, and barley
- Breakfast cereals
- Low-fat dairy products that may contain starch additives

These are foods that are likely to have high glycemic index (GI). The glycemic index (GI) is a scale of 0 to 100 that ranks foods according to the extent to which they raise blood-glucose levels in the blood relative to pure glucose, which ranks one hundred. We can find information about food and their corresponding GI values online.

We're advised to choose food with the lower GI option to control our blood-glucose levels.

While GI values provide good guidance, they are only a measurement to rate glucose absorption in the blood from ingesting food. It is not a measure of the sugar fructose. Pure fructose has a GI value in the range of ten, and it is not a healthy source of carbohydrates.

We also need to be aware that GI measures only the rate of absorption and not the absolute amount of carbohydrates in food. That is called the glycemic load (GL). GL takes into account both GI and the total number of carbohydrates in a serving of a particular food.

Overeating low GI foods with high GL can also lead to excess fat storage due to excess carbohydrates in the diet—although that is not as bad as consuming foods with a high GI and GL.

The best way to deal with all this is to analyze available lists of GI and GL values of the food you commonly eat. Make your own lists of foods you should eat more of and those you will abstain from.

Choose rightly. Avoid all processed, refined carbohydrates, starchy vegetables, and grains.

Get your carbohydrates instead from green leafy vegetables, roots, stems, and flowers of vegetables—with the exception of starchy tubers like potatoes, mushrooms, whole fruits, nuts, seeds, and legumes.

It would be best to consume minimally processed vegetables—either raw or lightly cooked. Fresh vegetables contain life-enhancing micronutrients that keep better with less cooking.

Soluble fiber in plant tissue forms tiny barriers that hinder digestive enzymes in the small intestines from getting to the carbohydrates. This process slows down digestion and the absorption of sugars into the

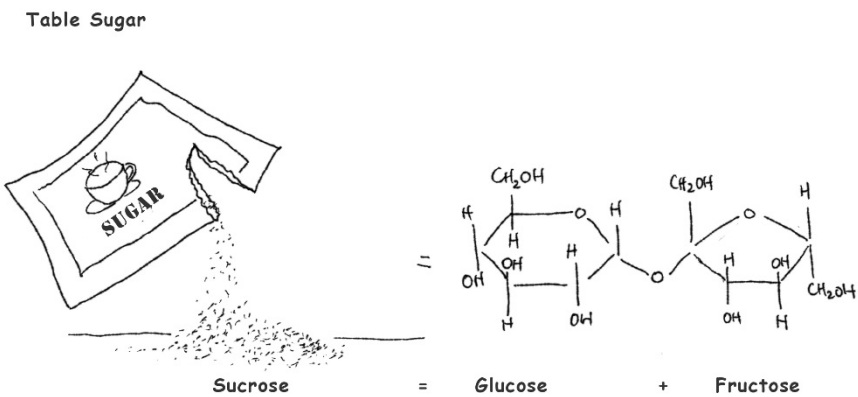
bloodstream (lower Glycemic Index).

Incidentally, the fibers in whole grains don't reduce the GI because these non-soluble fibers don't form effective barriers against digestive enzymes.

Get your carbohydrates from the right food sources.

How many carbohydrates one should eat depends on one's physical activity level, since most excess calories from carbohydrates get stored as fat.

Is Sugar a Poison?



When we talk about sugar, the additive we use to sweeten everything, we're referring to a molecule called sucrose. Sucrose is a glucose-fructose disaccharide. When we consume sugar, free glucose and fructose enter the bloodstream.

Glucose, we know, is a molecule all cells in the body use for energy. The hormone insulin regulates it in the blood. Fructose doesn't affect the secretion of insulin and can only be processed in the liver. Therefore, the GI of sucrose is in the range of 60 percent.

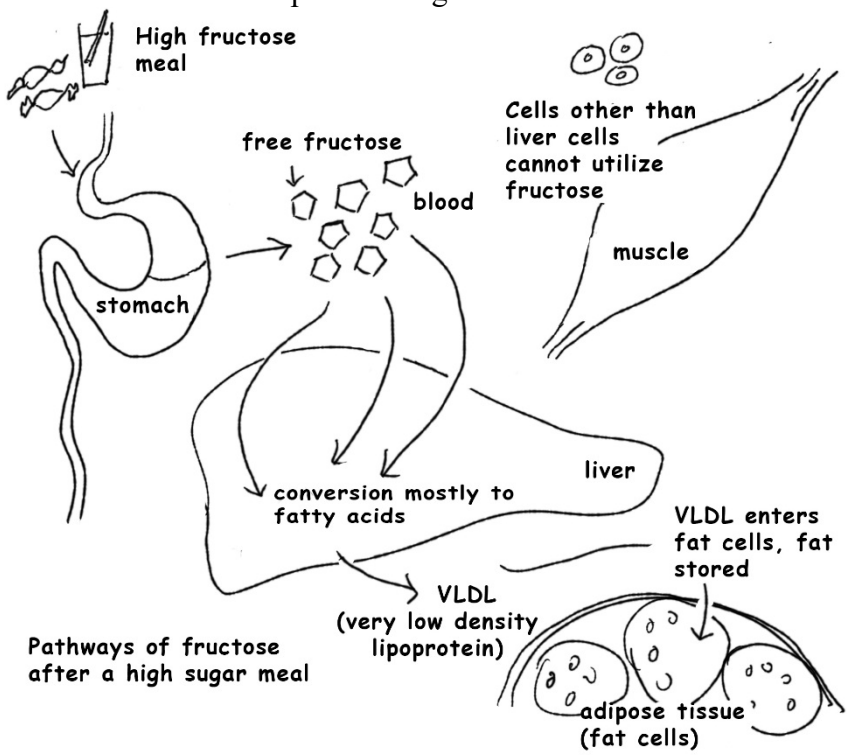
The metabolic pathway of fructose is similar to that of alcohol. Both can be processed only by the liver.

If we compare the above diagram with the processing of glucose, we observe the difference:

While some of the absorbed glucose gets used up by other cells in

the body and even stored in muscles, fructose gets used only when it enters the liver. This means that all fructose entering the bloodstream needs to be processed by liver cells, to remove it from the body.

Fructose level is not moderated in the body. No hormone like insulin is secreted in response to high-fructose blood levels.



Pathways of fructose after a high sugar meal

What are the health repercussions of long-term overconsumption of fructose?

- Leptin resistance. Leptin is a hormone that gives us the sensation of satiety, which means a sensation of being not hungry anymore. It normally kicks in twenty minutes after the start of a meal. Fructose has been known to suppress the effects of leptin, thereby causing us to keep eating.
- Obesity, since the body converts excess fructose to fat.
- Type 2 diabetes, metabolic syndrome, whereby the body cannot regulate sugar in the blood. While fructose doesn't trigger insulin secretion in the body, some have hypothesized that it is a cause of insulin resistance, a condition where the cells in the body cannot respond to insulin anymore, leading to

metabolic syndrome.

- Cardiovascular disease and stroke, since excess fructose exits the liver as VLDL (Very Low Density Lipoprotein). VLDL is a precursor to more LDL (Low Density Lipoprotein), which are the components of plaques that get deposited in the arteries.
- Damage to tissue by glycation. This is most observable in skin and bone damage. The symptoms are akin to premature aging, because the cellular structures in the body get damaged in the process.
- Gout due to increased uric acid (a waste-product of fructose processing in the liver) deposited from the metabolism of fructose
- Nonalcoholic fatty liver disease. Overconsumption of fructose will overload the functions of the liver, thereby causing symptoms of fatty liver and liver tissue damage.

Sugar addiction poses the same health risks as alcoholism. The mechanics by which the body metabolizes alcohol (in its digested state, it is called ethanol) are similar to the way it metabolizes fructose. The key difference between consuming fructose and consuming alcohol is that fructose doesn't bring the sensation of drunkenness. Just because fructose doesn't intoxicate, put us on a high, or cause us to be merry, throw up, pass out, or get a hangover, that doesn't make it safer to consume than alcohol. Since there is no warning of an overdose, fructose is actually a more harmful substance.

Like alcohol, fructose is also very addictive. That is why it is very difficult for most of us who are used to having sugary drinks, sweets, chocolates, and other snacks to completely quit eating them.

Large quantities of fructose in our diets come from the following:

- Table sugar, soft drinks, sweetened drinks
- Fruit juices
- Baked goods,
- Confectionery
- Desserts
- Jams and preserves
- Sauces
- Natural sweeteners like honey and maple syrup
- Processed foods, sauces, and salad dressings
- Breakfast cereals
- Sweetened milk products
- Dried fruit

- Fresh whole fruit – fruit in its natural form, cut or pureed but not juiced, sweetened or extracted in any way.

Of all the items in the above list, whole fresh or dried fruit is the only safe source of fructose we can eat in abundance. Soluble fiber in the fruit hinders the absorption of fructose in the small intestines, slowing down the process to a safe level. Fruits also have micronutrients important for body functions, making them a reasonable payoff. Even so, we should consume fruit only in its whole form, not juiced, with moderation.

If you want to experience a 180-degree turn in your health for the better, see immediate loss of excess fat in your body, and alleviate aging-related disease symptoms, cut down on your sugar intake for good.

- Drink nothing that contains added sugar, such as soft drinks, energy drinks, and sweetened drinks.
- Drink no juices.
- Eat fruit whole in moderation.
- Eat very little amounts of desserts, cakes, or confectionery.

Stay away from any processed food that has added sugar, fructose, glucose syrup, or high-fructose corn syrup.

- Avoid natural sweeteners like honey, maple syrup, treacle, and so forth.
- Avoid dairy products that contain sugar.

If you find that cutting back on sweets and fructose-containing foods is very restrictive and difficult to do, this is only because sugar is addictive. The best way to stop addiction is to understand that things will get easier if you just stick to the process of breaking the habit. You can do the following:

- Go to your kitchen and discard all sources of sugar. It is all right to throw away food items that are more harmful than nutritious. Grab a garbage bag and toss away all sugar, chocolate mixes, candy, frozen muffins, ketchup, and so forth.
- Make a list of all the foods you will not buy from the grocery store anymore. That is easy enough. Then make sure you don't go shopping on an empty stomach.
- Think of all the healthy foods you should eat instead and stock the kitchen only with those. If you're used to munching on potato chips in front of the TV, buy a bag of oven-roasted mixed nuts instead. That is not a perfect solution, but it is a good start to wean you off refined carbohydrate snacks.

If you failed to meet your goals and binge on a sweet or two, don't despair. This is a lifelong habit change. Think positively and get back on the right track.

How to Keep Excess Fat Off by Eating Right

Fat is an excellent fuel source for most tissues in the body. The breakdown of fatty acid molecules by cells releases a chemical called ATP (Adenosine Tri-phosphate). Production of ATP and its role in providing energy to cells is a complex process. All we need to understand now is that fat breakdown provides the cells with ATP which is energy for activities such as:

- Movement (in muscles)
- Heat production
- Biochemical functions like cell growth, wound healing, and so forth

Apart from the brain, which needs glucose for fuel, most cells in the body can burn fat for energy. When we have enough fuel, we will feel energetic. When our fuel source runs low, however, we will automatically feel lethargic and hungry.

Most of us today have excess fat stored in our adipose tissues that lie under the skin. This being the case, we should feel energetic, but we don't. Some of us even feel hungry all the time, when in theory the energy in stored fat should keep us going for hours, even days.

We feel deprived of energy because the cells in our bodies that should burn fat are not burning fat effectively. This is caused by the lingering presence of insulin in the blood.

Insulin, by acting to clear up glucose from blood, signals all cells to prefer glucose as a fuel source. When cells start metabolizing glucose instead of metabolizing fat, fat gets locked up in the adipose tissues and made unavailable as fuel for the metabolism.

Another way to understand this is to imagine two switches one can flip so cells get energy: the glucose-burning switch and the fat-burning switch. Only one switch is allowed to be turned on at a time. Normally the fat-burning switch is turned on. Fat is an innate fuel of choice for most cells because it is a form of fuel the body can store in good amounts. Glucose, on the other hand, runs out quickly because it is stored in very limited amounts as glycogen in the liver and muscles.

If glucose levels are high via regular high-carbohydrate diets, the

glucose-metabolizing switch is turned on, and the fat-metabolizing switch is turned off due to the influence of insulin.

The process of turning switches on and off takes weeks. Though it sounds like a simple idea, it is actually a complex regulatory process whereby genes are activated to produce enzymes for the respective fuel-utilizing machinery.

Control the amount of carbohydrate-rich foods in your diet to relieve your body from having to deal with the constant onslaught of glucose in the blood. This will result in metabolic changes in which the body will readjust its fuel choice naturally—to burn fat instead of glucose. When you change your diet this way, it will take about one hundred days to realize an improvement in your fat-utilizing capability.

The other effect of insulin, as mentioned earlier, is that it converts excess sugars into fat. This further increases fat stored in fat cells. If the body is switched on to burn glucose—while excess glucose that enters the blood is stored as fat—more fat gets stored up and locked away in adipose tissues. When glucose is converted to fat, there is no more fuel source for the cells that want only glucose as fuel. The result is a feeling of fatigue and hunger. When we get fatigued, we move less. When hunger strikes and we consume more carbohydrate-rich foods, the fat-storing cycle continues.

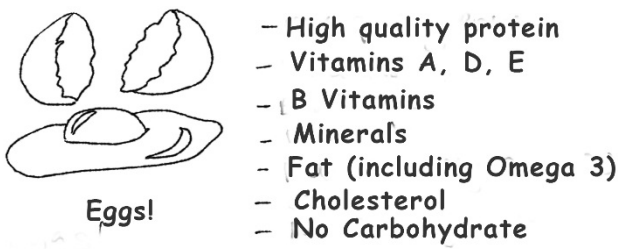
Listen to your body. Are you hungry for carbohydrates? Do you suffer lack of energy between meals? Do you suffer hunger swings? Are you putting on fat despite exercise and diets? Take refined carbohydrates out of your diet. The beginning of a new diet low in refined carbohydrates might be an uncomfortable experience, with feelings of fatigue and mood swings. The body needs a bit of time to change the switch—from glucose metabolizing to fat metabolizing, to unlock fat stores and to burn fat. Keep it up. Be patient and hang in there.

So, What's For Breakfast?

The traditional breakfast menu for most of us in the developed world is dominated by refined carbohydrates. The first meal of our day usually centers on breads, cereals, pancakes, pastries, sugary dairy drinks, fruit juices, and so forth. These foods, consumed in abundance, will ultimately enter the blood as simple sugars. This triggers the secretion of insulin, followed by a cascade of biochemical events that

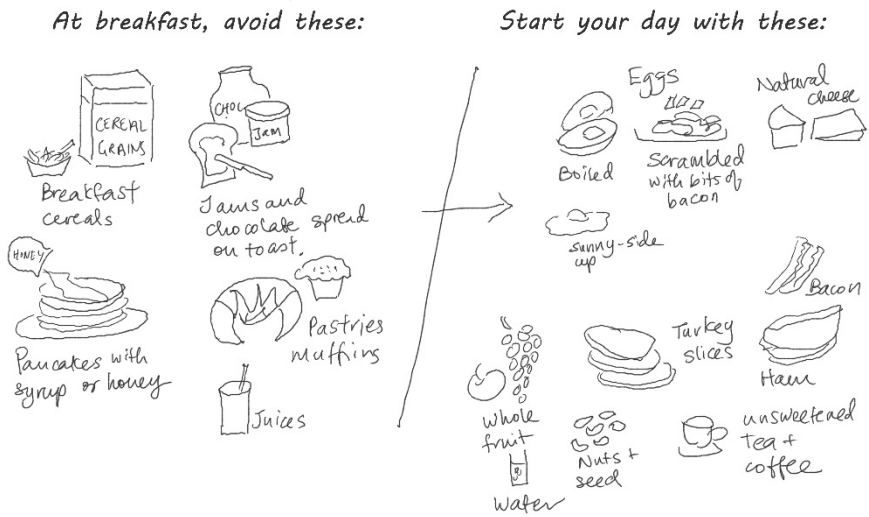
manifest in the short term as hunger swings and in the long term as fat accumulation. Reduce your intake of refined carbohydrate foods by starting each day with more nutritious, low-carbohydrate alternatives such as the following:

- Eggs—boiled, poached, served sunny side up, scrambled, served in omelet form
- Vegetables—non-starchy types, mushrooms
- Strips of lean chicken, beef, pork or venison, spiced with a thin strip of bacon
- Cheese (the natural, full-fat variety)
- Mixed nuts and seeds (raw or dry roasted is best)
- Avocado
- A serving of whole fruit



Eggs? This sounds like the opposite of our conventional idea of a "healthy low-fat breakfast," right? Yes it is, because low fat is not the healthy option. The idea that eating grain-based food is healthy is also a myth. Grains like wheat, barley, oats, and rice, including the whole grain varieties, are high in easily digestible carbohydrates. Consuming stacks of bread, pancakes, and rice porridge will just send your blood glucose levels up.

If you must eat grain-based foods for breakfast, be aware of the effect they will have on your metabolism. Reduce the portions of these foods and replace them with low-carbohydrate alternatives.



You don't need to eat less, just eat differently.

Try this yourself for a month. You will notice a difference in the way your body looks and feels.

Water and the Fountain of Youth

Mentioning the dietary importance of water is almost a cliché. We all know this, but many of us still don't drink enough. It could be that water drinking isn't interesting and can sometimes be inconvenient. Water is our most important dietary component because almost 70 percent of our body mass is water.

Water has these functions:

- Water creates bulk in all living cells. Lack of water (or dehydration) causes cells to lose their mass, get deformed, and eventually die. As a result, tissues and organs get damaged.
- Water is the universal solvent. Water dissolves gases, mineral ions (salts), vitamins, toxins, and other organic and inorganic substances. These chemicals and biochemicals need to be dissolved in water for essential chemical interactions to occur.
- Water is a transporter. Large organic molecules—for example, lipoproteins, enzymes, hormones, neurotransmitters, and cells like

blood cells—need to move around the body. Moving liquid in the body is predominantly water. It is like a river that needs to stay fluid. When dry, the river will not allow boats carrying cargo to pass from one location to the next.

- Water maintains the internal environment of the body in a process called homeostasis (*homo* means "the same" or "constant", and *stasis* means "to stand still"). Homeostasis is a process where the fluids in the body is always kept within a fixed conditions.

Tissues and organs are very sensitive to change in its environmental parameters, which include body temperature, acidity, electrolytes and solutes, sugar, blood pressure, and so forth. These parameters have to be kept constant. The cardiovascular system keeps blood and body fluids in constant flow so that the regulatory measure initiated by an organ like the liver is transported to the whole body.

Water facilitates cell-to-cell communication. Cells in different organs and tissues communicate via neurotransmission and hormone secretion. These communication processes require fluids in the body.

How Much Water is Enough?

Your daily water requirement depends on several factors: for example, the climate you're in, your activity level, the food you eat. As with everything else, there is no need to get number bound and start counting the number of glasses to drink per day. It is just best to drink little sips of water throughout the day. Have a glass before and after each meal. Carry a bottle of water around with you every time you engage in physical activity. Drink more when the day is hot or when you have to engage in physical activity. Drink before you even get thirsty. Drink plain, clean water. It doesn't matter if it is sparkling or still, piping hot or on the rocks. Have it with a drop of fresh lemon if it helps you drink enough.

Water activity is a scale to measure the amount of available water in food. Pure water has the water activity value of 1. Anything that is dissolved in water is called a solute. Solutes like sugar, salt, alcohol, and other nutrients added to water reduce the water activity.

This means that a food or beverage with more solute dissolved in it has less function as a source of water than pure water.

When choosing a thirst-quenching, body-fluid-replenishing beverage, it is better to drink plain water or drinks with very few ingredients dissolved in them, such as lightly brewed unsweetened

teas and some brands of diet beverages.

Juices, milk, sweetened drinks, soft drinks, energy drinks, alcoholic beverages, and soups don't count as good sources of water because of their high-solute content.

Fresh, plain water is ultimately the ideal source of water to stay hydrated.

Drink No Calories

There are two benefits to sticking with this rule:

1. You get more hydration from your beverage (as discussed in the previous section).
2. You avoid consumption of excess calories, which are more likely to be empty calories from refined carbohydrates.

This rule alone can help reduce most people's daily calorie intake by a quarter, if not more.

Most of us don't realize how many calories we consume when we drink juice, non-diet soft drinks, sweetened coffees and teas, lattes and other milk-based drinks. Not following this "Drink No Calories" rule often leads to the following:

- Overconsumption of sugar
- Overconsumption of calories
- Overeating caused by hunger swings, which are due to sugar consumption.

Foods in liquid form tend to go down the hatch quickly and mindlessly since there is no need to scoop, bite, and chew. When we drink our food, most of the time we're not even aware of the quantity we consume. Often we drink more of calorie-rich, sweetened beverage than we would water because the low-water activity of the beverage actually leaves us thirsty for more.

Drinking alcoholic beverages is best kept a recreational activity. Alcohol is intoxicating, and we all know the consequences of overindulgence. Treat it like chocolate; indulge in a little every once in a while, but don't make it a staple.

About 99 percent of the time, drink no calories.

Drinking no calories leaves us with limited choices on what we can drink most of the time:

- Water—still, sparkling, with a dash of lemon or herb

- Coffee and tea—black and unsweetened
- Diet soda

This is one of the easiest rules to stick to. It may also be, for some individuals, the most effective measure for quick calorie reduction and perhaps excess fat loss. *So let's drink water to health!*

"No disease that can be treated by diet should be treated with any means." Maimonides

The Micro-nutrients

- **Minerals**

Most of our dry body weight constitutes minerals. These minerals are deposited in bones. Bones provide the frame that protects the organs and supports muscles to enable mobility.

One thing we have to understand is that bones are dynamic. Minerals move into and out of bone tissue all the time. We need to replenish these minerals through proper nutrition—by consuming foods rich in calcium, phosphorous, and other salts for bone building.

Other minerals found in the body are dissolved in the body fluids. This process aids in biochemical reactions that are essential to life.

Salts like sodium, potassium, and chloride are involved in the electrical conduction in nerve and muscle cells.

Trace minerals like iodine, manganese, zinc, and selenium are essential for optimal biochemical processes.

Minerals are found in whole foods. A diet based on meat and vegetables is certain to be adequate in mineral content.

- **Vitamins**

Vitamins are essential in our diets because the human body cannot produce them. Vitamins are organic compounds that are necessary for biochemical reactions. Some of these vitamins, such as vitamins A, C, and E, are known for their protective antioxidant activity.

Vitamins are chemically and biochemically very diverse and can be separated into two broad categories—water soluble and fat soluble. It is good to be mindful of this characteristic of vitamins if we want to be sure we get enough of them in our diets.

Water-soluble vitamins are vitamins C and the various vitamin Bs.

Oil-soluble vitamins are vitamins A, D, E, and K.

Literature on vitamins, minerals, and phytochemicals—as well as

their sources and benefits—are available online and in books. There is generally no need for us to consume vitamin supplements. A diet that is complete with fish, meats, eggs, and fresh vegetables would contain sufficient amounts of micronutrients needed for a healthy existence.

Diet Myths of the Twenty-First Century

The last three decades have seen the largest increase in obesity rates among populations in the developed world. This is despite the fact that there is more awareness and more information about it.

Part of the problem is that the information given to us over the last decades was not completely accurate. Some of the worst advice we got are the myths, which many of today's writers of good books on nutrition and weight loss have tried to debunk.

The best way to discuss diet is to first focus on nutrition. Good nutrition comes only from a diet of good-quality food packed with the necessary macronutrients and micronutrients. Good-quality food keeps the body healthy. A healthy body can do wonders. It is full of energy; it can do things, replenish itself, and stay young.

The opposite is true for poor diets.

Eating poorly actually accelerates the aging process more quickly than lacking exercise.

Diet Myths to Debunk Forever

There are numerous diet myths that need to be debunked. These myths have misled us into decades of misinformation and the obesity epidemic. There are many theories as to how the myths have come to be so ingrained in our beliefs. Our concern here is to identify the most damaging of diet myths and debunk them.

Myth 1: Whole grains are healthy.
Truth: As mentioned earlier, whole grains are, like refined grains, empty sources of carbohydrates. These cause insulin levels to rise just as badly as refined carbohydrates and starchy vegetables.

Myth 2: Fat in food causes obesity.
Truth: Uncontrolled carbohydrate consumption, not the fat in food, that leads to hunger swings causes overeating and obesity. Good fats

in food have essential function, while refined carbohydrates only get stored up as fat.

Myth 3: A vegetarian diet is healthier.
Truth: A vegetarian diet gives no benefit because it is difficult to get adequate amounts of protein from vegetable sources. It is also difficult to get a good variety of food to ensure there is no deficiency in any nutrient.
Unless one is very well versed in vegetarian nutrition (or if one has motives other than health to go vegetarian), it is best not to adopt a strict vegetarian diet.

Myth 4: Egg yolks are bad because they contain cholesterol.
Truth: Egg yolks contain cholesterol, yes. Egg yolks are also a cheap source of vital nutrients. Furthermore, cholesterol is not a harmful substance. Cholesterol molecules are found in our cell membranes that keep cell structure flexible and functional. The body regulates its own production of cholesterol with cholesterol in the diet.

Myth 5: Cholesterol causes heart disease.
Truth: The cholesterol we're led to believe is the "bad cholesterol" that causes heart disease isn't even cholesterol. It is a large molecule called "type-B low-density lipoprotein" (LDL).
Type-B LDL levels in the blood are controlled by reducing visceral fat accumulation (fat around the waist) by embarking on a low, controlled carbohydrate diet and exercise.

If your meals are predominantly made up of refined carbohydrates and you find yourself needing a snack not two hours later, it's time to change the menu. We can see that the bulk of problems we face today are associated with diets consisting of predominantly refined carbohydrates. It is sensible to ignore myths and focus on eliminating the real dietary culprits.

The Dangers of Poor Diet

The human body is not formed as static material but is in constant flux. Cells continually get damaged due to age and wear, and need replacement. This regenerative process depends on an adequate supply

of macronutrients and micronutrients derived from the food we eat.

Nutrients are the building blocks of cell growth. Excessive or long-term calorie-deficient and/or nutrient-deficient diets deprive the body of its ability to regenerate itself, leading to breakdowns in tissue integrity and organ function. This deterioration happens gradually. The symptoms would be chronic in nature and will show up as signs of aging; fatigue; poor mental state; bone, muscle, and organ degeneration; cardiovascular disease; and a compromised immune system leading to infectious ailments and even cancer due to a weak immune system.

Malnutrition is a detriment to health. The irony is that while we, people in this modern world, think we have too much to eat, the reality is that most of us don't get enough of the essential nutrients in our diets to maintain good form.

LONG-TERM CALORIE DEFICIENCY

Calorie-deficient diets include excessive fasting, skipping meals, eating less, constantly going hungry, excessive calorie counting, going on a perpetual low-calorie diet, and struggling with anorexia. Most people go on a calorie deficit with the aim of losing weight. The premise is that if you eat fewer calories than you burned, you will lose weight.

The energy deficit brings about loss in fat for only the first couple of weeks; then it stops. Some people even find themselves putting the pounds back on.

A long-term calorie deficit puts the body into a starvation mode. The resting metabolic rate (RMR) goes down to conserve energy for survival.

This is a natural survival mechanism of human physiology, meant to sustain us in time of famine, which in Paleolithic times didn't last too long.

In this mode, fat is spared, defeating the dieter's purpose for going on the diet in the first place. More threatening to health is that biological processes considered nonessential for immediate survival, such as cell rejuvenation, muscle growth, bone replacement, and neurological and immune function, will slow to a halt. When this happens, everything slowly breaks down. This is the precursor to aging and aging-related diseases. Hence this method of dieting is not the solution to getting forever young and fit.

NUTRITION DEFICIENCY

Nutrition-deficient diets come in many forms. They include low-fat or no-fat diets, low-carbohydrate diets, and vegetarian diets.

Deliberately denying the body of any nutrient is like taking away the building blocks for growth and replenishment. It is the surest way to kick-start deterioration of the organs and accelerate the aging process.

Fat, protein, and carbohydrates all have a role in protecting the body from aging and illness. Nutrient-deficient diets have no place in our aim to stay young and fit.

MINDLESS EATING

Eating anything on hand, with the tendency to consume calorie-dense processed food, may seem like the opposite of calorie- and nutrient-deficient diets, but there is a similarity—it's a cause of malnutrition. Processed food, fast food, junk food, and the like are calorie dense and nutrient empty. This means that a small quantity of food can easily make up one's daily calorie needs without fulfilling the nutrient needs.

The high calories consumed with this kind of diet come predominantly from refined carbohydrates. As explained in the section on insulin, excessive refined-carbohydrate intake will cause obesity and metabolic disorder, leading to chronic diseases. Contrary to popular understanding, being fat doesn't mean being well nourished. You can put on fat and still have a body starving for vital nutrients.

EMOTIONAL EATING

Foods with addictive ingredients, such as sugar, carbohydrates, and alcohol, have calming and pleasurable effects on us. When we consume food to justify a poor emotional state, we're participating in emotional eating. Like taking narcotics and depression drugs, eating to feel better has only a transient effect. Emotional eating often leads to gross overeating and obesity.

Stop emotional eating by being aware of it and getting the appropriate help in curing the poor emotional state.

The Young-and-Fit Diet

Eat for replenishment and growth. Consider your nutrition mindfully. This is the philosophy for a diet practice that would keep you young and fit for life. A good attitude toward eating right has to do with the following attributes:

- **Mindfulness**

Be aware of everything you eat. Anything you put into your system during meals or outside of mealtime constitutes your diet. Being mindful helps us break nasty habits that add on empty calories. Habits include nibbling on snacks, tasting morsels of food offered at supermarkets, finishing up your kid's leftovers, eating free bread sticks at restaurants, and so forth.

- **The Macro-nutrient Shift**

Get enough protein and fat in your diet. These macro-nutrients should make the bulk of your daily meal. Plan your meals around protein-containing foods. Supplement your meals with good-quality dietary fats.

- **Selective Carbohydrate Sources**

Control your carbohydrates. These should come from non-starchy vegetables, nuts, and seeds. Avoid grains, starchy vegetables, and processed foods that contain sugar.

- **Quality of Choice**

Eat only quality food for micronutrients. The fresher the meats, fruits and vegetables, and so forth, the closer these foods are to their original state. Therefore, the more vitamins, phytochemicals, and minerals you will get from them.

- **Wholesomeness**

Choose whole foods over processed food. Processing depletes food of its natural micronutrients. Chemical additives added for the shelf-life extension of food need to be detoxified by the liver. This puts unnecessary stress on the organ.

- **Variety**

Eat a variety of food. This is a good habit to ensure you get all the nutrients the body needs. It is also a safeguard against accumulation of chemicals or an overdose of certain micronutrients in the body.

Eating to stay young and fit is only about getting the right amount of nutrients for the purpose of building and replenishing tissues and organs.

Forget Calorie Counting

"A good decision is based on knowledge and not on numbers."
Plato

A calorie is a unit measure of energy. The food "calories" we see on food labels are calculated in a physics laboratory by measuring the amount of heat emitted when a food sample is ignited in a bomb calorimeter. Corrections in the readings are sometimes made with consideration to the digestibility and absorption of particular food types in an attempt to reflect more closely the biological processing of the food.

The reality is different. The human body doesn't function like a bomb calorimeter. We don't combust our food. Different sets of enzymes in different tissues metabolize fats, proteins, and carbohydrates, and different hormonal mechanisms regulate them.

Our digestive system also absorbs different foods differently. Foods that are highly processed get digested, and their nutrients are absorbed into the blood more efficiently than whole, mildly cooked foods.

There are so many variables in the digestion, absorption, and metabolism of food that it is impossible for science to accurately calculate the "calories" in food products.

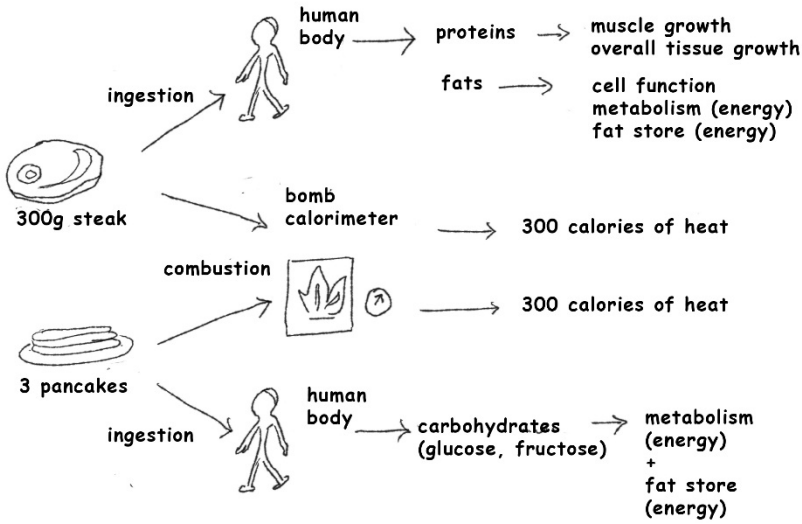
It is, therefore, better not to get number bound. Calories mean almost nothing in the reality of food and nutrition. We can use the calorie scale to estimate how much we eat. This estimation will not indicate whether the food we eat will keep us healthy or make us fat.

For example, calories from a piece of steak will have a different effect on your body than the same amount of calories from a pancake. The nutrients absorbed from the two sources of food will also have different functions.

Steak is protein rich and almost carbohydrate free. Consumption of steak releases amino acids in the blood. There will be an insignificant blood-sugar spike and very little insulin production.

A stack of pancakes, on the other hand, is carbohydrate rich. Eating pancakes will boost blood-sugar levels and lead to an insulin response.

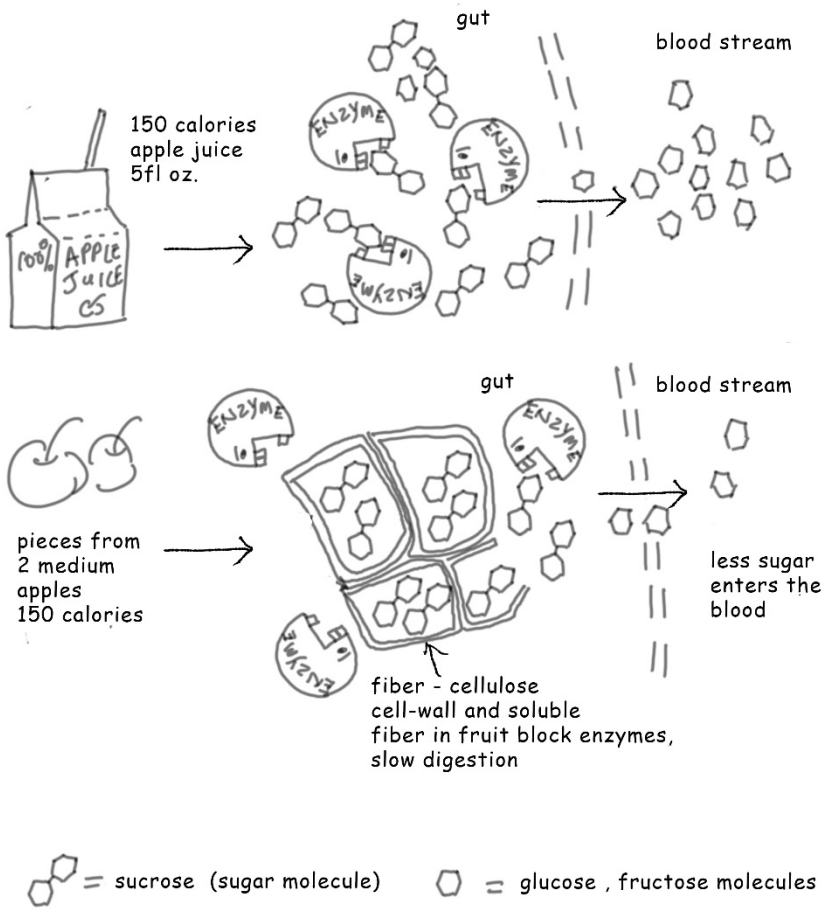
This is an example of how two meals that measure the same number of calories in a calorimeter affect different hormonal responses. Depending on other dietary parameters and an individual's physical state, the rise in insulin levels from eating pancakes poses a greater potential for immediate fat storage than a similar calorie-containing meal of steak.



Both servings of meat and pancakes might burn in a caloriemeter to liberate 300 calories of energy, but in the human body, both are processed differently and used for different biochemical function.

Here is another example. Calories from apple juice will create a metabolic response in the body that is different from the same amount of calories from whole cut apple pieces. While the nutrients of both apple juice and whole apples are the same, the physical structures of the two foods are different. In whole fruit, nutrients are packed in cells and fiber, which hamper the digestion process of its nutrients.

The nutrients from whole apples get into the blood at a much lower rate and less efficient manner than apple juice because of the soluble and insoluble fibers still present in whole apples. Drinking apple juice causes a quick rise in blood glucose and fructose levels—hence a higher insulin response. Eating whole apple pieces results in a slower absorption of sugars into the bloodstream. This process gives more time for the sugars to be metabolized, requiring less insulin involvement.



If knowing the number of calories doesn't help us stay healthy or lean, what is the point of counting calories in the first place? There isn't. As stated in the chapter on exercise, this is the same reason why we should not be too concerned about counting calories.

We don't get fat just because we eat too many calories. We accumulate fat in our bodies because we eat calories from the wrong *food source.*

The good news is that we don't need to eat less. We just need to eat differently. A diet consistently dominated by carbohydrates (especially refined carbohydrates) stimulates insulin production. The body becomes a fat-accumulating, fat-sparing, and carbohydrate-

burning machine. This is the diet prevalent in almost every part of the developed world. This is the diet of today's obesity epidemic.

To solve these constant obesity concerns, immediately make these changes to the diet:

- Reduce the carbohydrates in your diet, while consuming more protein and fat.
- Replace all refined carbohydrates you normally consume at mealtime with fresh meat, fish, poultry, non-starchy vegetables, some fruit, legumes, nuts, seeds, mushrooms, and eggs.
- Avoid foods containing high quantities of fructose and alcohol. Enjoy these but only in homeopathic quantities.

A change in diet—to one that is very low in refined carbohydrates, sugars, and alcohol; and is rich in good-quality protein, fat, and fresh vegetables—will cause a good change in the metabolic machinery. With a low incidence of blood sugar surges leading to insulin spikes, fat burning will start to kick in. It may take a couple of weeks for this change to happen, but it will happen. When the cells in the body start to burn fat, fat loss becomes a long-term reality.

Healthiness and Not Thinness

There is no harm in having a little fat here and there or now and then. We need to focus on being healthy, not thin. Good health is possible only if your body is fed right. A healthy body is a young body.

The bonus is that eating right raises your metabolism and improves your energy levels. This will almost automatically work in your favor to reduce excess body fat.

Simple Meal Planning

Planning your meals is a good way to keep yourself on a clean diet. Meal planning isn't about making spreadsheet tables so you have to plan what you want to eat days in advance. That can be complicated and impossible to follow through on. It is more effective to take a look at every aspect of the meal—the where, when, and how—rather than to focus only on the what. Here are some ideas:

- Have meals only at the dining table, not in front of the television or at the work desk.

- Focus on what you eat while you eat. When you're aware of your eating, you're less likely to overeat. Refrain from eating on the move or in front of distraction like the television.
- Start every meal with the right kind of food. Fill your belly first with lean meats, eggs, and vegetables before going for anything else.
- Snack wisely. Nuts—especially raw, lightly roasted, and unsalted ones—are great. Fruit, vegetable sticks, boiled eggs, and some cheese can also make good snacks. Snacks being snacks, they should be eaten mindfully.
- Drink water before and after every meal to feel satiated more quickly.
- Eat slowly. About twenty minutes need to pass before your body realizes you have had enough.
- Learn to stop eating when you're not hungry anymore. This takes a little practice. Be conscious of the differences between feeling hungry, feeling not hungry anymore, feeling full, and feeling stuffed.
- Don't completely deprive yourself from the foods you love; however, reduce your exposure to them. Do enjoy them in small amounts, as a treat every once in a while.

Healthy Eating as a Lifestyle

"Going on a diet" is something you may do on the short term. Most diets are unsustainable. They are either too depressing to follow or unhealthy. True healthy eating for staying young and fit is a permanent lifestyle choice. That is why you shouldn't overanalyze, count, weigh, or make big diet charts.

All you have to do is understand that your reason for choosing what you eat is for the sole purpose of good body function, which is the prerequisite of staying young and fit. Experiment with different foods. Our bodies react differently to different eating patterns and habits. Understanding how your body works is the answer to managing your health.

Smart Grocery Shopping

Not everybody enjoys grocery shopping, and the food industry is aware of that. If it is not entertaining or interesting, we don't buy. This is how companies are able to genetically select food crops, modify, package, and color lesser-quality food and get more sales.

It is time to change our attitudes toward grocery shopping. Imagine a woman from Paleolithic times going out into the wilderness to gather daily staples. How would she have done it? Would she have selected the brightest-colored thing just because it looked interesting? No, because brightly colored plants and grubs in nature are likely to be poisonous. The woman would have picked the freshest things available and only those she had learned from previous generations to be edible and nutritious.

Would a caveman pick up a dead animal rather than hunt for a live beast just because it was easier to get? I would think not, because he probably understood that rotten flesh can be deadly.

Modern society is in love with processed food. Instead of being more discerning, we have forgotten the importance of nutrition for survival. Thinking about food quality can be boring, so we allow ourselves to be manipulated. To sell more, food processors add color,

shape, packaging, and all kinds of marketing slogans to entice us into picking out their produce at the grocery store.

When we thoughtlessly give convenience and packaging precedence over the nutritional quality of food, we choose wrong. When we choose wrong, we eat wrong.

Grocery Shopping Made Simple

A good way to get only quality nutrition through grocery shopping is to stock your cart with healthy items while leaving unhealthy food on the shelves. We can also follow this simple rule:

- Let 80 percent of food in your shopping cart contain raw, whole, unprocessed meats, fish, eggs, and non-starchy vegetables and fruit.
- Let 15 percent contain some cheese and delicatessen, nuts and seeds, butter, good oils, canned fish, coffee, and tea.
- Let 5 percent be anything else.

If you want to transform your life today, refrain from stocking up on foods that contain added sugar, refined carbohydrates, and alcohol. These are usually over-processed and ready to eat. Experience has shown that the more of these foods we keep at home, the more of them we consume.

Smart grocery shopping is the key to eating right at home.

Strategies for Dining Out

Eating establishments are in the business of making profits from the food they serve you. In order to achieve that, these businesses have to focus on everything else but serving healthy food. Some of the strategies restaurants use include the following:

- Stuffing the menu with cheap, refined carbohydrates so the dishes look bigger
- Over-seasoning, as in adding too much salt, MSG, and other artificial flavor ingredients to mask lesser-quality cooking or poorer-quality food
- Using trans fats for frying, because the oil can be re-used in the fryer much longer than healthier fats
- Deep-frying or frying everything with a lot of fat because it

cooks the food more quickly

- Adding starch and sugar to food, especially sauces, gravies, and salad dressings for a more attractive look and taste
- Serving huge portions to look like more value for the money. This causes the consumer to overeat.

For most of us, dining out is no longer something we do only on special occasions. It has become a necessity. Some of us even dine out every day.

Apart from the obviously unhealthy fare, most restaurants also serve healthy alternatives. Many restaurants also take special requests from their customers. We, therefore, have a choice of choosing food that is right for us while dining out. All we need is a strategy:

Before Ordering Food

- If you know the restaurant menu already, plan in advance what you can eat there. If you don't know the restaurant or are unsure what is on the menu, try not to go there in a ravenous state. Have a small healthy snack beforehand.
- Your beverage order should ideally be limited to water, diet soda (if you have to), unsweetened tea, or coffee.
- Send back the free bread, chips, nuts, or whatever between-meal snack is provided.

What to Order

- Pick the dish with a larger serving of whole lean protein. It could be a whole chicken breast, a steak, fish, pork, beef, venison, lamb, or eggs.
- Get a good serving of vegetables. These should be mainly greens, cruciferous vegetables, stems and roots, non-starchy vegetables, and vegetables cooked as lightly as possible without too much sauce and seasoning. Salads and steamed vegetables are normally good choices.
- If the dish comes with a serving of fries, pasta, or rice, ask if you could substitute them with more vegetables.
- Ask for sauce on the side. Sauces and gravies normally contain sugar, starch, fat, and too much salt. Having them on the side allows you to control how much of that you eat. No ketchup please!

How to Eat

- Eat the meat and vegetables first before consuming the other components of the dish. Undigested proteins and fiber in the stomach tend to satisfy hunger quickly, making you less likely

to binge on the carbohydrates.

- Eat slowly.
- Stop eating when you're no longer hungry.
- Drink water between mouthfuls.
- Leave as much of the potatoes, bread, pasta, and rice—the carbohydrate-rich foods—behind as possible. Remember that carbohydrates are cheap stuffing restaurants put on plate to give consumers the impression they are getting more value for their money. We know that consuming refined carbohydrates is the cause for a host of health issues. Understand that leaving them behind is not a waste. They are simply not worth eating.
- Skip dessert or—maybe, why not?—share your dessert.
- Have black coffee or tea without sugar.
- Eating out will not be a healthy-diet-busting activity. Plan ahead what you will and will not eat before entering a restaurant. If you're invited to a restaurant that doesn't offer healthy options in its menu, have a healthy snack beforehand so you don't go with an empty stomach.

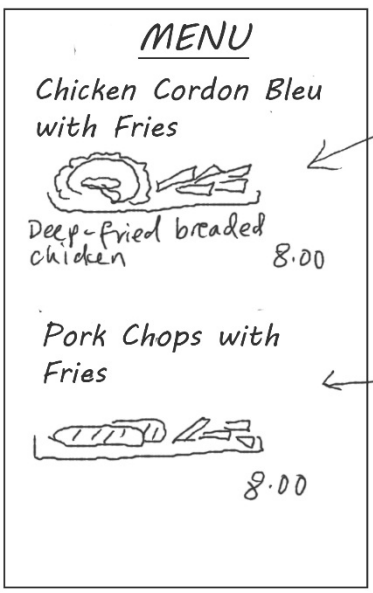
Enjoy Your Special Occasions

Just because you're on the path of a healthy lifestyle doesn't mean you cannot enjoy fun foods on special occasions. It is good to forget restrictions every once in a while and give yourself a treat. Your body can take incidental onslaughts of naughty food.

If your lifestyle is generally about eating clean, you can afford to party. This is actually good for you. The process of staying forever young, forever fit is about replenishment. Enjoyment with friends and family over fun food sends pleasure hormones into your body. Your body will de-stress, and your immune system will get a boost, even if you have that champagne and four servings of chocolate cake.

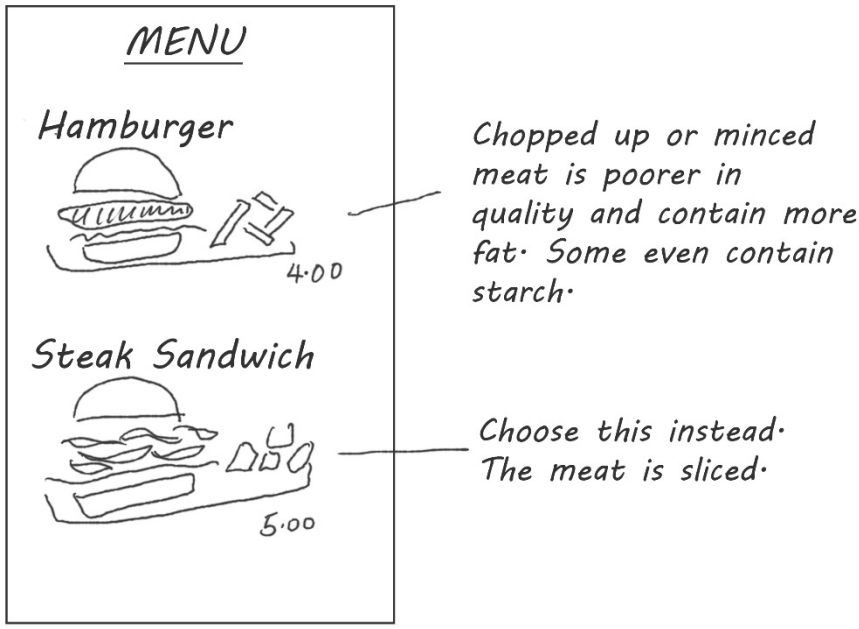
Dish dominated by carbohydrate is not a good choice

Choose instead the dish with larger serving of lean protein.



Deep-fried meat is not a good choice.

Better to have meat that is not deep-fried but grilled. Do ask for vegetables instead of French-fries, or leave the fries un-eaten.



Diet is a Big Part of Staying Young and Fit

This is a big chapter because nutrition affects the state of our health more than anything else, even physical exercise. Everything you eat affects your physiology sooner or later.
Eat well to live well. Eat mindfully with the aim of getting the essential macronutrients and micronutrients.

Understand your own body. Try different foods and observe the effect food has on your health. You will see a great transformation in your physical form by adjusting your diet in the right direction.

Once you've got the hang of eating well, grocery shopping, cooking, and dining out will be a pleasure. There will be no confusion, doubts, or the feeling of overeating—or the idea that you need to abstain from anything enjoyable.

Maintaining healthy eating habits is a lifestyle anyone can adopt by following certain rules. Adhere to these rules as closely as possible. They do get easier with perseverance. There are many ways to

motivate ourselves to eat well. For some of us, the obvious positive effects of our clean diet are motivation enough. There are also books to read on proper nutrition, and even self-hypnosis recordings that can sort out any cause for eating poorly.

Do whatever it takes to keep on track. Try to understand what works for you to stay forever young and fit.